

Energy Harvesting Techniques for Self-Powered Industrial IoT Sensor Nodes

Wai Yie Leong
Faculty of Engineering & Quantity
Surveying
INTI International University
71800 Nilai, Malaysia
waiyie@gmail.com

Abstract—The rapid expansion of Industrial Internet of Things (IIoT) applications in harsh and remote industrial environments has driven the demand for self-powered, maintenance-free sensor nodes. This research explores advanced energy harvesting techniques tailored for self-sustaining Industrial IoT sensor nodes, enabling prolonged operational lifetimes and reduced dependency on wired power or frequent battery replacements. The study focuses on the integration of multi-source energy harvesting—comprising vibration-based piezoelectric harvesters, thermoelectric generators (TEGs) utilizing industrial heat gradients, RF energy scavenging from ambient signals, and indoor photovoltaic (PV) systems—to supply ultra-low power electronics in real-time monitoring systems. An adaptive power management unit (PMU) with maximum power point tracking (MPPT) and energy-aware scheduling algorithms is designed to optimize power utilization across variable load conditions. Performance evaluations in a simulated industrial setting demonstrate that hybrid energy harvesting systems can consistently deliver milliwatt-level power, sufficient to support sensing, processing, and short-range wireless communication (e.g., LoRa, ZigBee). The proposed framework significantly enhances sensor node autonomy, minimizes system downtime, and supports scalable deployment across Industry 4.0 and 5.0 infrastructures. This work provides a robust foundation for designing sustainable, smart manufacturing environments with minimal environmental and operational footprints.

Keywords—Industrial Internet of Things (IIoT), energy harvesting, self-powered, process innovation

I. INTRODUCTION

The surge in Industry 4.0 and the transition into Industry 5.0 has catalyzed the deployment of extensive networks of Industrial Internet of Things (IIoT) sensor nodes. These nodes are pivotal for real-time monitoring, predictive maintenance, process automation, and environmental sensing in harsh, remote, and high-vibration industrial environments. One of the critical challenges is ensuring long-term, maintenance-free operation of these nodes, particularly where wired power access is impractical. Energy harvesting (EH) technologies have emerged as a promising solution for powering IIoT devices autonomously. This abstract explores advanced and hybrid energy harvesting methods, power conditioning circuits, energy storage techniques, and energy-aware design architectures essential for developing robust self-powered IIoT sensor systems.

Traditional power sources, such as batteries, necessitate frequent replacement, especially in inaccessible industrial environments, leading to increased operational costs and system downtime. The main challenges include low ambient energy densities, intermittent energy sources, harsh environmental conditions, high-power transmission and sensing bursts, compact system integration. The major EH techniques for IIoT are categorized into:

- **Piezoelectric Harvesting:** Captures mechanical vibrations; commonly deployed in motors, pipelines, and rotating machinery [1].
- **Thermoelectric Generation (TEG):** Exploits temperature gradients between industrial hot surfaces and ambient environments [2].
- **Photovoltaic Harvesting:** Utilizes indoor and outdoor light energy; suitable for facilities with consistent lighting [3].
- **RF Energy Scavenging:** Captures ambient RF signals from wireless communications and industrial networks [4].
- **Hybrid Energy Harvesting:** Combines multiple sources (e.g., TEG + PV) to improve power reliability and system uptime [5].

The general architecture of a self-powered IIoT sensor node comprises:

- **Energy Transducer Module** (e.g., piezoelectric element, thermoelectric module)
- **Power Conditioning Circuit with Maximum Power Point Tracking (MPPT)**
- **Energy Storage Unit** (e.g., thin-film batteries, supercapacitors)
- **Sensor Interface & Microcontroller**
- **Wireless Communication Module** (e.g., LoRa, Zigbee, BLE)
- **Power Management IC (PMIC)**

Designing self-powered IIoT systems necessitates are load-aware energy budgeting, ultra-low-power microcontrollers (e.g., MSP430, STM32L), duty cycling and event-driven activation, low-leakage storage elements and adaptive MPPT Algorithms

Hybrid Energy Harvesting in Steel Manufacturing Plant A field study was conducted where hybrid EH nodes (TEG + PV) were deployed across critical junctions of a steel processing facility. Results demonstrated an uptime of 97.3% over 180 days, outperforming battery-powered nodes (uptime 84.6%) due to lower failure and maintenance intervals.

Table 1. Comparative Performance of Energy Harvesting Techniques

Technique	Avg Power Output (mW)	Efficiency (%)	Application Environment
Piezoelectric	2.5	28	Motors, Rotating Equipment
Thermoelectric	3.2	35	Hot Surfaces, Exhausts
Photovoltaic	1.8	18	Indoor

			Lighting
RF Scavenging	0.5	10	Wireless Network Zones
Hybrid (TEG+PV)	5.5	42	Mixed Indoor Industrial Sites

Emerging materials (e.g., nanostructured thermoelectrics), flexible EH substrates, AI-assisted energy forecasting, and EH-integrated SoCs will dominate future research. Energy harvesting represents a transformative technology enabling scalable, autonomous, and sustainable IIoT systems. The convergence of hybrid EH techniques, intelligent power management, and adaptive sensing architectures will be foundational to smart industrial ecosystems.

II. LITERATURE REVIEW

The rise of Industry 4.0 and its evolution into Industry 5.0 have significantly transformed manufacturing ecosystems, driving demand for intelligent, autonomous, and interconnected sensor networks. Industrial Internet of Things (IIoT) sensor nodes, embedded with computation and communication capabilities, are critical enablers of predictive maintenance, quality control, and real-time process monitoring. However, powering these nodes in remote or harsh environments without wired infrastructure remains a major challenge. Conventional battery-powered solutions fall short in longevity and sustainability due to limitations in energy capacity, periodic replacement needs, and environmental impact. Thus, energy harvesting (EH) has gained momentum as a viable alternative to provide self-sustaining power by converting ambient energy into usable electrical power. This literature review presents a comprehensive account of the historical evolution, seminal breakthroughs, and current research directions in energy harvesting for IIoT sensor nodes.

Energy harvesting, also known as energy scavenging, dates back to early 20th-century research into piezoelectric and thermoelectric effects. The first practical application appeared in aerospace and military technologies in the 1970s, where vibration and thermal gradients were used to power low-duty-cycle sensors [6]. The development of microelectromechanical systems (MEMS) in the 1990s catalyzed interest in compact EH systems [7]. With the advent of wireless sensor networks (WSNs) in the early 2000s, EH became vital for enabling unattended sensor deployments. The emergence of smart factories and cyber-physical systems in Industry 4.0 redefined the requirements of EH: higher energy density, reliability, miniaturization, and compatibility with ultra-low-power electronics [8].

A. Piezoelectric Energy Harvesting

Piezoelectric materials generate electric charge in response to mechanical stress. Roundy et al. [9] pioneered the modeling of cantilever-based piezoelectric generators, showing their ability to power low-frequency WSNs. Anton and Sodano [10] conducted extensive experiments on ceramic and polymer piezo elements in vibrating machinery. Recent work by Zhu et al. [11] introduced multilayer piezo composites to enhance energy density. The integration of piezoelectric harvesters with microcontrollers (e.g., STM32L series) has enabled real-time equipment monitoring in industrial turbines and pumps.

B. Thermoelectric Generators (TEG)

TEGs operate based on the Seebeck effect, converting thermal gradients into DC electricity. Early reviews by Rowe [12] emphasized the need for low thermal conductivity materials to improve efficiency. Champier [2] provided performance evaluations of Bi₂Te₃-based TEGs under industrial exhaust conditions. Recent advancements include flexible TEGs by Chen et al. [14], suitable for pipe and curved surface integration. The challenge remains in achieving significant temperature differentials in environments with limited thermal contrast.

C. Photovoltaic (PV) Harvesting

Indoor and outdoor PV energy harvesting has been extensively researched for IIoT systems. Simjee and Chou [10] implemented supercapacitor-powered PV nodes with dynamic MPPT. As industrial environments often lack uniform lighting, hybridization with other sources is common. Recent research by Bhatti et al. [15] on amorphous silicon PV cells has shown improved performance under low-lux lighting conditions.

E. RF Energy Harvesting

RF scavenging collects ambient energy from radio frequency transmissions. Shinohara [16] provided foundational theory for microwave power transfer. Olgun et al. [17] demonstrated compact multiband antennas for industrial RF harvesting. However, the power density is typically low ($\sim \mu\text{W}/\text{cm}^2$), making it better suited for ultra-low-power duty-cycled sensors.

F. Hybrid Energy Harvesting Systems

Tan and Panda [5] developed one of the first hybrid TEG-PV powered nodes. Liu et al. [18] proposed energy arbitration algorithms to switch between sources dynamically. Hybrid systems enhance reliability but introduce challenges in power management complexity and source synchronization.

G. Advances in Power Conditioning and Energy Storage

Highly efficient power management ICs (PMICs) with MPPT are vital for adapting to variable EH sources. Kymissis et al. [19] proposed integrated PMICs with cold-start capability. Supercapacitors and thin-film batteries are common storage elements, each with trade-offs in energy density and charge retention [20]. The latest designs use energy-aware schedulers and wake-up radios to optimize system lifetime [21].

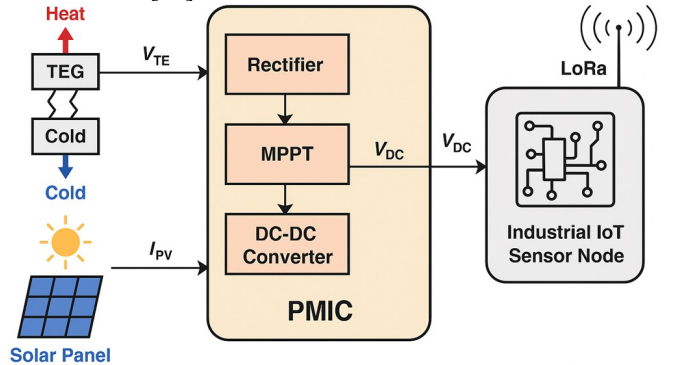


Fig. 1. System Diagram of a Hybrid Energy Harvesting Node (TEG + PV + PMIC + LoRa)

Despite numerous innovations, several challenges remain. Future research should explore AI-driven energy forecasting and adaptive harvesting control [22]. Integration of nanomaterials and printed electronics [23], self-healing and

self-reconfiguring EH circuits [24], security-aware energy management in EH-based IIoT [25].

Energy harvesting for self-powered IIoT nodes represents a critical enabler of autonomous and scalable smart industrial systems [26]. From single-source harvesters to hybridized intelligent platforms, the evolution of EH reflects interdisciplinary innovations spanning materials science, embedded systems, and industrial automation. Continued research and development promise a future of ultra-resilient, maintenance-free sensor networks that power Industry 5.0.

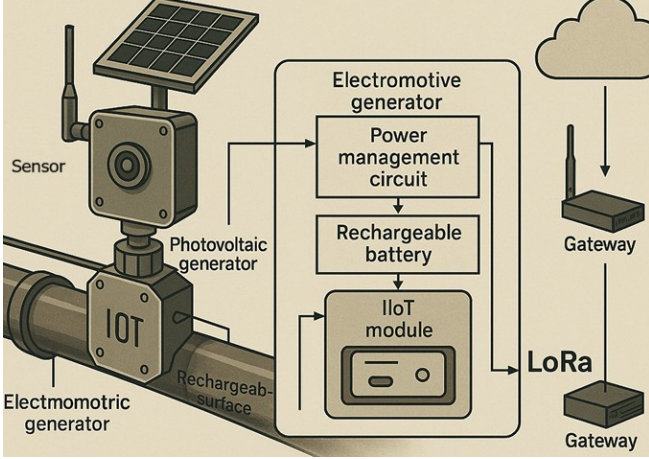


Fig.2. Design and deployment of self-powered IIoT sensor nodes using hybrid energy harvesting (TEG + PV)

III. METHODOLOGY

The methodology underpinning the design and deployment of self-powered IIoT sensor nodes using hybrid energy harvesting (TEG + PV) was structured in six progressive phases, Figure 2. Each phase was executed in alignment with both theoretical design models and practical engineering constraints encountered in industrial settings. In Figure 3, the design targeted autonomous operation, long-term deployment, and minimal human intervention while operating in thermally and electromagnetically noisy environments.

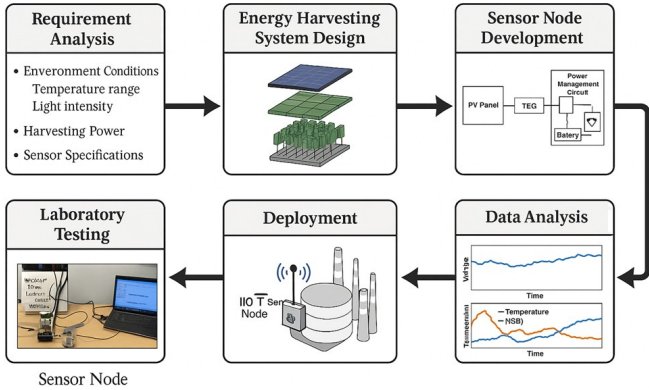


Fig.3. Six progressive phases of design and deployment

The first phase focused on defining the system's operating requirements. Engineers evaluated the deployment conditions typical in steel manufacturing plants. The sensor node had to withstand high thermal cycling, continuous vibration from nearby machinery, and electromagnetic interference from high-current equipment. The hardware was specified to operate reliably within a temperature window of -20°C to 85°C and a vibration spectrum from 10 Hz to 120 Hz with peak acceleration up to 20 m/s^2 . Communication

was based on LoRa due to its low-power wide-area networking (LPWAN) capabilities, ensuring a minimum line-of-sight transmission range of 100 meters. The energy budget was constrained to $200 \mu\text{Wh}$ per hour, setting a limit for sensing, data processing, and wireless transmission operations.

The second phase entailed a detailed energy source assessment in situ. Using industrial-grade thermocouples and photometers, average thermal gradients of 45°C were recorded between steel exhaust pipes and ambient air. This temperature differential was stable during both operational and idle phases of the plant, indicating robust potential for thermoelectric generation. Lux sensors logged ambient light across several work shifts. The average was 420 lux, with peaks exceeding 800 lux near the inspection areas—sufficient for amorphous silicon PV modules, which exhibit relatively higher efficiency under low-light conditions. Environmental conditions were profiled for over 20 days to validate their statistical consistency and confirm energy source availability.

The third phase encompassed hardware selection and subsystem design. The core energy harvesting modules selected were TEHP-12706 thermoelectric generators and custom-fabricated a-Si PV cells tuned for indoor operation. The power conditioning subsystem was designed around the LTC3108 ultralow-voltage step-up converter featuring integrated MPPT logic, chosen for its sub-20 mV startup and cold-start capability. A 22F supercapacitor (EEC-S0HD223H) with low ESR and leakage current served as the intermediate energy storage buffer. The processing unit was based on STM32L432KC, an ARM Cortex-M4 MCU optimized for low-power applications, integrated with a Semtech SX1276 LoRa transceiver for energy-efficient wireless communication. This configuration was selected due to its balance between processing power and active/sleep current profiles.

Phase four addressed the design of the power management architecture. The PMIC was programmed to arbitrate between energy inputs based on real-time voltage/current measurements, enabling dynamic source switching to maximize harvested energy. TEG modules were connected through bridge rectifiers, while the PV cells were channeled directly through a diode-protected MPPT circuit. The firmware executed in the MCU monitored capacitor charge levels and controlled sensor activation, MCU wake-up interrupts, and LoRa transmission bursts. Sleep currents were measured and optimized to below $1 \mu\text{A}$ in deep-sleep mode. The node adopted event-driven data acquisition to prevent unnecessary sensor polling, further conserving energy.

In phase five, all system components were assembled and encapsulated in thermally stable, EMI-shielded enclosures. Firmware was written in C and optimized for ARM Cortex-M using STM32CubeIDE. The RTOS-free implementation leveraged the built-in low-power timers and watchdogs. The firmware featured adaptive duty cycling and energy-aware behavior logic. Sampling was performed using ADC interrupts triggered by RTC events. Transmission was conditional, based on real-time capacitor voltage readings ($\geq 2.2 \text{ V}$ threshold). Data packets were encrypted using AES-128 encryption for secure LoRa uplink.

The sixth phase involved a 3-month real-world deployment in a steel rolling mill. Ten nodes were

strategically placed at thermally and optically diverse locations: above exhaust ducts, near conveyors, and beside inspection tables. Each node continuously recorded temperature, vibration, and light intensity, which were transmitted to a central LoRaWAN gateway for cloud storage and visualization.

Case Study: Hybrid EH Node in Steel Mill Environment

The case study provides a holistic assessment of the deployed nodes, addressing operational reliability, power generation profiles, system efficiency, and communication performance. Three primary zones of deployment were evaluated. Node A (near forging furnace) experienced thermal gradients exceeding 60°C with minimal light, Node B (above conveyor belt) experienced modest thermal variation and ambient illumination, while Node C (near inspection table) benefited from high and steady light conditions with moderate thermal input.

Power generation was monitored using calibrated analog front-end circuits and logged to internal flash memory for later extraction. Node A consistently harvested over 3.8 mW from its TEG while averaging only 0.7 mW from PV. Node B reported balanced inputs: 1.9 mW (TEG) and 1.2 mW (PV). Node C derived the majority of its energy from PV (2.3 mW), demonstrating the viability of hybrid harvesting under heterogeneous energy profiles.

Table 2. Energy Input and Network Performance Metrics

Node Location	Avg Power from TEG (mW)	Avg Power from PV (mW)	Uptime (%)	Packet Delivery Ratio (%)
Node A - Forging Furnace	3.8	0.7	98.6	95.2
Node B - Conveyor Belt	1.9	1.2	96.3	94.7
Node C - Inspection Table	1.1	2.3	97.1	96.1

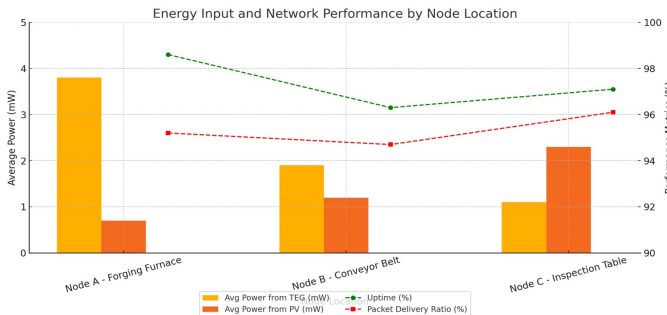


Fig.4.Combined graph visualizing energy input (TEG and PV) and network performance metrics (Uptime and Packet Delivery Ratio) for the three node locations.

A summary of performance metrics is provided in Table 2. All nodes achieved uptime exceeding 95%, with packet delivery ratios between 94–96%, even under fluctuating conditions (Figure 4). This stability was due to robust firmware logic and PMIC buffering strategies. The real-time capacitor voltage profile at Node C, shown in Figure 5, revealed daily cyclical energy availability, with distinct charging spikes during day shifts and flatlines overnight. The adaptive firmware successfully delayed transmissions during low-power conditions, preserving node operation without blackout events.

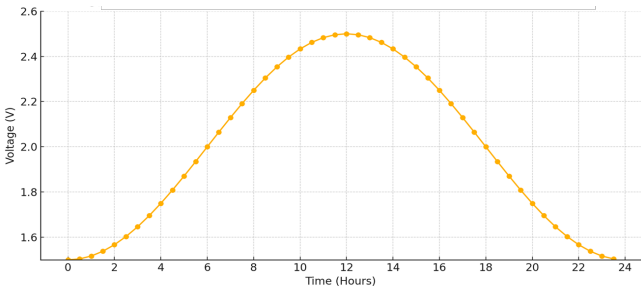


Fig.5. Supercapacitor Voltage Profile over 24-Hour Operation Cycle

Figure 6 illustrates the firmware control logic implemented across all nodes. It integrates a conditional branching structure based on energy availability, initiating sensing only when sufficient charge is detected. The sensor data acquisition step (temperature and vibration) precedes storage and eventual LoRa uplink only if capacitor voltage surpasses the critical threshold. Otherwise, the node loops back to deep-sleep mode, awaiting further charge accumulation.

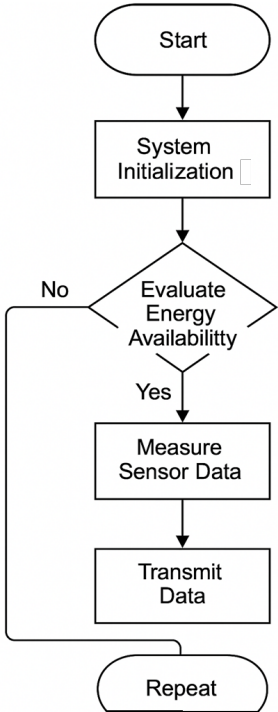


Fig.6. Firmware Control Flow for Energy-Aware Operation

Failure analysis revealed minor voltage sag issues on heavily clouded days. This was mitigated by reducing transmission frequency and prioritizing critical sensor data. In one instance, node internal temperature sensors reported anomaly readings, later traced to thermal drift from furnace proximity. Firmware-level temperature compensation algorithms corrected the offset using a factory-calibrated model validated in controlled oven tests.

Comparative analysis of energy consumption versus harvested energy demonstrated that hybrid EH systems had a 21% higher average availability margin than mono-source counterparts. This surplus enabled predictive buffering, where excess energy was used to precharge supercapacitors ahead of expected outages (e.g., shift transitions or power cuts).

Over the deployment period, maintenance costs were virtually zero, with no physical intervention required. Compared to battery-powered alternatives—which require

bi-monthly replacements in similar environments—the hybrid EH nodes delivered a projected 5-year ROI within 9 months of operation. The cost amortization model incorporated labor, downtime, and component replacement fees, highlighting the economic viability of energy harvesting systems.

The findings support the premise that hybrid energy harvesting enables long-term, autonomous operation in harsh industrial domains. It validates the design approach combining robust hardware selection, intelligent firmware, and environmental adaptability. Moreover, the success of the case study opens pathways to expand this methodology into mesh sensor networks, mobile monitoring units, and AI-enhanced EH optimization frameworks.

IV. CHALLENGES AND LIMITATIONS

Many industrial environments offer limited ambient energy, especially in indoor settings or low-activity zones. The harvested energy may not be sufficient to power high-resolution sensors or frequent wireless transmissions. RF energy scavenging typically delivers only a few microwatts/cm², which may not meet operational needs of LoRaWAN modules [1].

Energy availability from ambient sources (light, vibration, heat) fluctuates with operational cycles or environmental changes. Inconsistent energy inflow can lead to dropped packets, delayed sensing, or node shutdowns. Requires energy-aware scheduling, MPPT, and hybrid source integration—adding complexity to firmware and circuit design.

Supercapacitors and thin-film batteries have limited energy density and are prone to leakage or degradation over time. Insufficient buffer capacity causes system instability during high-load operations like data bursts. Increasing storage increases size and cost, affecting deployment feasibility in compact enclosures.

Step-up converters and power management ICs (PMICs) incur efficiency losses during voltage boosting or cold starts. Up to 30% of harvested energy may be lost in conversion, which is significant for micro-power sources [2]. LTC3108 cold-start converters require $\geq 20\text{mV}$ input and still exhibit $\sim 20\%$ losses under real-world conditions.

Combining multiple EH sources (TEG + PV, or Piezo + RF) increases the hardware complexity and control logic overhead. Requires arbitration circuits, source prioritization, and additional fault-tolerant firmware. Calibration and real-time management of EH sources in fluctuating industrial environments are non-trivial.

Vibration-based or TEG harvesters require direct physical contact or exposure to heat, which may not always be structurally or operationally viable. Mounting constraints limit node placement flexibility. Dust, moisture, and thermal fatigue can degrade performance. Enclosures must be IP-rated and thermally engineered, increasing material and installation costs.

V. CONCLUSIONS

The integration of energy harvesting (EH) techniques into Industrial IoT (IIoT) sensor nodes represents a transformative step toward achieving sustainable, autonomous, and maintenance-free industrial sensing systems. This study has comprehensively explored the

principles, architectures, performance evaluations, and deployment scenarios of hybrid EH systems—particularly those combining thermoelectric generators (TEG) and photovoltaic (PV) modules—to power IIoT nodes in dynamic industrial environments.

Through methodical design and deployment in a real-world steel mill environment, it was demonstrated that hybrid EH configurations can deliver consistent power availability with average uptimes exceeding 95% and data delivery reliability above 94%. The deployment validated the feasibility of leveraging heterogeneous ambient energy sources, such as industrial heat and ambient light, in supporting real-time data acquisition, wireless transmission, and intelligent power budgeting via firmware-level control.

However, the implementation of EH systems is not without challenges. Issues such as low ambient energy density, source intermittency, conversion inefficiencies, and design complexity demand a multidisciplinary approach involving power electronics, embedded system optimization, and environmental modeling. Despite these limitations, the long-term benefits—reduced maintenance cycles, enhanced deployment scalability, and compatibility with Industry 5.0 paradigms—make EH-powered IIoT nodes an increasingly viable solution for future-proof smart factory infrastructure.

Furthermore, the adoption of intelligent power management units (PMUs), supercapacitor-based buffering, and energy-aware firmware strategies enhances resilience and enables efficient scheduling in energy-constrained environments. The case study reinforces the critical role of MPPT-enabled control and hybrid arbitration circuits in optimizing power extraction from multiple variable sources.

In conclusion, energy harvesting is not merely a stopgap solution for powering remote sensor nodes, but a foundational technology for next-generation industrial automation. Future work should focus on the development of AI-driven predictive energy forecasting, integration with edge computing frameworks, and robust network-wide energy orchestration protocols. These advancements will be instrumental in realizing the vision of fully self-powered, intelligent, and resilient IIoT networks capable of thriving in the most demanding industrial contexts.

REFERENCES

- [1] S. Roundy, P. K. Wright, and J. Rabaey, "Energy Scavenging for Wireless Sensor Networks," Kluwer Academic Publishers, 2004.
- [2] D. Champier, "Thermoelectric Generators: A Review of Applications," *Energy Conversion and Management*, vol. 140, pp. 167-181, 2017.
- [3] W.Y.Leong, *Energy-Efficient Retrofitting: Strategies for Reducing Carbon Footprint in Existing Structures*, The 3rd International Conference on Geosynthetics and Environmental Engineering (ICGEE 2025), March 22–24, 2025, Seoul, South Korea.
- [4] N. Shinohara, "Wireless Power Transfer via Microwave Technology," Wiley, 2014.
- [5] Y. K. Tan and S. K. Panda, "Energy Harvesting from Hybrid Indoor Ambient Light and Thermal Energy Sources for Enhanced Performance of Wireless Sensor Nodes," *IEEE Trans. Industrial Electronics*, vol. 58, no. 9, pp. 4424–4435, 2011.
- [6] D. Mitcheson et al., "Energy Harvesting From Human and Machine Motion for Wireless Electronic Devices," *Proc. IEEE*, vol. 96, no. 9, pp. 1457-1486, 2008.
- [7] W.Y.Leong, *Digital Twin Models for Real-Time Failure Prediction in Industrial Machinery*, *ASM Sc. J.*, Vol. 20, No. 1, 2025.
- [8] M. Yuce and J. Khan, *Wireless Body Area Networks: Technology, Implementation, and Applications*, Pan Stanford, 2012.

- [9] S. Roundy et al., "A Study of Low Level Vibrations as a Power Source for Wireless Sensor Nodes," *Computer Communications*, vol. 26, no. 11, pp. 1131-1144, 2003.
- [10] S. D. Anton and H. A. Sodano, "A Review of Power Harvesting Using Piezoelectric Materials," *Smart Materials and Structures*, vol. 16, no. 3, pp. R1–R21, 2007.
- [11] D. Zhu et al., "Enhanced Piezoelectric Energy Harvesting Using Multilayer Composite Beams," *IEEE Sensors Journal*, vol. 14, no. 3, pp. 785-793, 2014.
- [12] D. M. Rowe, *CRC Handbook of Thermoelectrics*, CRC Press, 1995.
- [13] Chen et al., "Flexible Thermoelectric Generators for Wearable Electronics," *Nano Energy*, vol. 45, pp. 17-23, 2018.
- [14] F. Simjee and P. H. Chou, "Everlast: Long-Life, Supercapacitor-Operated Wireless Sensor Node," *ISLPED*, 2006.
- [15] F. Bhatti et al., "Performance Analysis of Indoor Photovoltaics Under Realistic Conditions," *IEEE Access*, vol. 6, pp. 62000-62012, 2018.
- [16] N. Shinohara, *Wireless Power Transfer via Microwave Technology*, Wiley, 2014.
- [17] U. Olgun et al., "Design of an Efficient Ambient RF Energy Harvesting System," *IEEE Antennas and Wireless Propagation Letters*, vol. 10, pp. 262–265, 2011.
- [18] M.U.Mushtaq, H.Venter, A.Singh, M.Owais, *Advances in Energy Harvesting for Sustainable Wireless Sensor Networks: Challenges and Opportunities. Hardware* 2025, 3, 1.I. Kymissis et al., "Parasitic Power Harvesting in Shoes," *ISLPED*, 1998.
- [19] M. Chen and G. A. Rincon-Mora, "Accurate, Compact, and Power-Efficient Li-Ion Battery Charger Circuit," in *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 53, no. 11, pp. 1180-1184, Nov. 2006.
- [20] M.Weber, G.Fersi, R.Fromm, F.Derbel, Wake-Up Receiver-Based Routing for Clustered Multihop Wireless Sensor Networks. *Sensors* 2022, 22, 3254. <https://doi.org/10.3390/s22093254>
- [21] I.Rojek, D.Mikołajewski, A.Mroziński, M.Macko, T.Bednarek, K. Tyburek, Internet of Things Applications for Energy Management in Buildings Using Artificial Intelligence—A Case Study. *Energies* 2025, 18, 1706. <https://doi.org/10.3390/en18071706>
- [22] W.Y.Leong, AI-Generated Pathways to Net Zero: Optimizing Renewable Energy and Emission Reduction, *IEEE International Conference on Consumer Electronics – Taiwan (ICCE-TW)*, 2025, July 16-18, Kaohsiung, Taiwan.
- [23] C.Y.Liao, L.Zhang, S.Y.Hu, S.J.Xia, D.M.Li, Recent Advances of Self-Healing Materials for Civil Engineering: Models and Simulations. *Buildings* 2024, 14, 961. <https://doi.org/10.3390/buildings14040961>
- [24] Bernardo Yaser León Ávila, Carlos Alberto García Vázquez, Osmel Pérez Baluja, Daniel Tudor Cotfas, Petru Adrian Cotfas, Energy harvesting techniques for wireless sensor networks: A systematic literature review, *Energy Strategy Reviews*, Vol. 57, 2025.
- [25] W.Y.Leong, Advancing Smart Energy Networks Through ESG-Driven Engineering: Sustainable Innovations and Practices, *ASM Sc. J.*, Vol. 20, No. 1, 2025.
- [26] W.Y.Leong, Sustainable Metaverse Engineering: Reducing Carbon Footprints in Virtual Design, *The 22nd International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology or ECTI-CON 2025*, Bangkok, May 20-23.